SANDIPAN-DEY

exercise5.py

```python
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics.pairwise import cosine_similarity

# Collect user input
documents = []
text = input("Enter document contents or type `exit` to end
while text.strip() != 'exit':
    documents.append(text.strip())
    text = input("Enter document contents or type `exit` to
print("The cosine similarities between these documents are:

# WRITE YOUR CODE HERE
vectorizer = CountVectorizer(stop_words='english', binary=T
X = vectorizer.fit_transform(documents)
print( cosine_similarity(X) )
```

100% (15:30)

Terminal

```
codio@horizonliquid-elasticsusan:~/workspace$ python3 exercise5.py
Enter document contents or type `exit` to end input: This is the first
Enter document contents or type `exit` to end input: This document is t
Enter document contents or type `exit` to end input: exit
The cosine similarities between these documents are:
[[1.         0.70710678]
 [0.70710678 1.        ]]
codio@horizonliquid-elasticsusan:~/workspace$
```

Guide

Collapse

Coding Exercise 5

In the top-left pane, write code that will calculate and print the Cosine Similarity of the `documents` which will be captured by keyboard input.

Make sure to:

- Use `CountVectorizer`
- Filter out punctuation
- Filter out stop words
- Ignore case
- Score by existence (i.e. `binary` scoring)

Output the Cosine similarity.

Test your code using the terminal in the bottom-left:

TRY IT

Example Input/Output:

```
This is the first document.
This document is the second document.
exit

The cosine similarities between these documents are:
[[1.         0.70710678]
 [0.70710678 1.        ]]
```

Submit your code for feedback by pressing the button below.

The first step is to properly vectorize the documents:

```
vectorizer = CountVectorizer(stop_words
X = vectorizer.fit_transform(documents)
```

The second step is to calculate the cosine similarities between the documents and print them out:

```
print( cosine_similarity(X) )
```

Check It!

LAST RUN on 12/27/2023, 4:48:51 AM

Check 1 passed
Check 2 passed
Check 3 passed

Mark as Completed

Back to dashboard